



TRISH PURKAIT

📍 Subashgram, Kolkata - 700147 | ✉ trishpurkait@gmail.com | ☎ +91 6291616139
trishpurkait.github.io/Portfolio | github.com/trishpurkait | linkedin.com/in/trishpurkait | leetcode.com/u/trish0

SUMMARY

AI/ML and Generative AI-focused Computer Science and Engineering student skilled in Python, **Machine Learning**, **Deep Learning**, **NLP** and **RAG**. Experienced in building production-style AI applications with Flask, LangChain, Pinecone, Hugging Face and observability/evaluation workflows. Hackathon finalist with strong problem-solving, teamwork and quick-learning ability.

PROJECTS

- **MediQuery: Production-Style Conversational Medical RAG System** — Built a medical RAG chatbot using Flask, LangChain, Pinecone dense retrieval, BM25 sparse retrieval, cross-encoder reranking and session memory. Added Prometheus/Grafana monitoring, p50/p95 latency tracking, cost estimation, trace IDs, JSON logs and GitHub Actions quality gates. [[GitHub](#)]
- **LLaMA-3 E-Commerce Support Bot** — Fine-tuned Meta-Llama-3-8B-Instruct into a domain-specific customer support assistant using QLoRA/PEFT, 4-bit NF4 quantization and LoRA adapters. Trained on 44,884 retail support examples and served inference through a Gradio multi-turn chat interface. [[GitHub](#)]
- **Krishi Sohayak: Plant Disease & Crop Recommendation System** — Developed agriculture-focused ML modules: a CNN-based plant disease detector with Gemini-powered treatment recommendations and a Random Forest crop recommender that ranks top crops using soil and climate parameters with confidence scores. [[Plant GitHub](#)] | [[Crop GitHub](#)]
- **Deepfake Audio Detection using MFCC Features and Neural Networks** — Built a real-vs-fake audio detection workflow using Librosa feature extraction, MFCCs and neural network classifiers. Processed 22,859 audio samples with 48 extracted acoustic features for deepfake audio classification. [[GitHub](#)]

EDUCATION

- **Future Institute of Engineering & Management, Kolkata** — B.Tech in Computer Science & Engineering (2023-'26)
- SGPA: 7.91 (Till Now) | Core: Machine Learning, Deep Learning, Object-Oriented Programming
- **Central Calcutta Polytechnic, Kolkata** — Diploma in Computer Science and Technology (2020-2023)
- GPA: 8.8 | Core: Operating Systems, Data Structures & Algorithms, DBMS

TECHNICAL SKILLS

- **Programming:** Python, Java, C
- **ML/DL & NLP:** TensorFlow, Keras, scikit-learn, OpenCV, spaCy, Librosa, Pandas, NumPy, Matplotlib, Seaborn
- **Generative AI & RAG:** Hugging Face, LLMs, LoRA/QLoRA, LangChain, Pinecone, ChromaDB, Prompt Engineering
- **Web, Database & Tools:** Flask, HTML, CSS, JavaScript, MySQL, PHPMyAdmin, Git, GitHub, VS Code, Google Colab, Docker, Prometheus, Grafana, GitHub Actions
- **Problem Solving:** 75+ problems solved on LeetCode

ACHIEVEMENTS & CERTIFICATES

- FrostHacks 2025 - Finalist (Top 8 among 300+ teams)
- HackSphire 2024 - Selected in Top 30 among 200+ teams
- Data Science, AI & ML using Python - Ardent Computech Pvt. Ltd. (Dec 2023-Mar 2024)
- Deep Learning, GenAI & Prompt Engineering - Ardent Computech Pvt. Ltd. (Nov 2024-Mar 2025)

SOFT SKILLS & LANGUAGES

- Languages: English, Hindi, Bengali
- Soft Skills: Teamwork, Collaboration, Problem-solving, Adaptability, Creativity, Quick Learning